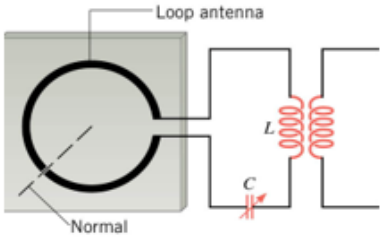
Worksheet # 6 Part A - Multiple Choice
Physics II

Name: _____ Date: _____

Multiple Choice Questions 1 though 10

1) What is the most efficient way for the antenna shown below to pick up Electromagnetic radiation?



A) The electric field oriented parallel to the normal to the antenna
B) The magnetic field oriented parallel to the normal to the antenna
C) Both the magnetic and electric field
D) The Electric field oriented perpendicular to the normal to the antenna
E) The magnetic field oriented perpendicular to the normal to the antenna

2) Mike goes to his favorite fast-food restaurant, he sees in the kitchen, hamburgers and French fries under a lamp to keep them warm, what type of electromagnetic waves does this lamp use to keep the food warm?

A) Visible light
B) Gamma radiation
C) Radio waves
D) Infrared radiation
E) Ultraviolet radiation

3) If it takes approximately 8 minutes for visible light to reach the earth from the sun, how would this time change for gamma rays?

A) Gamma rays would take a shorter time
B) Gamma rays would take the same amount of time
C) Gamma rays would take a longer time
D) Not enough information

4) The main difference between radio waves and visible light waves traveling in a vacuum is there

A) speed.
B) modulation.
C) frequency.
D) all of these.
E) A and C only.

5) For an Electromagnetic wave, how does the energy in the electric field compare to the energy in the magnetic field?

A) The electric field has the exact same amount of energy as the magnetic field.
B) The electric field has the more energy than the magnetic field.
C) The electric field has the less energy than the magnetic field.
D) The electric field has exactly half as much energy as the magnetic field.

6) If Mike starts at 1 m from a light bulb (assume point source) and walks to 3 m from the light bulb, how does the intensity of the light from the bulb change?

A) It is reduced by a factor of 1/3.
B) It is increased by a factor of 3.
C) It does not change.
D) It is increased by a factor of 9.
E) It is reduced by a factor of 1/9.

7) Which of the following objects is a blackbody emitter?

A) The Sun.
B) The Human body.
C) Ice.
D) A rock at room temperature.
E) All the above.
F) None of the above.

8) What part of the EM spectrum does the Sun emit most of its energy?

A) Visible light.
B) Infrared radiation.
C) Ultraviolet light.
D) Gamma Rays.
E) Radio waves.

9) How is the angle used to calculate the Intensity of light transmitted through a polarizer measured?

A) From the vertical to the electric field of the polarized light.
B) From the vertical to the transmission axis of the polarizer.
C) From the electric field of the polarized light to the transmission axis of the polarizer.
D) From the horizontal to the electric field of the polarized light.
E) From the horizontal to the transmission axis of the polarizer.

10) The blue sky is caused by what electro-magnetic wave effect.

A) Diffraction
B) Rayleigh Scattering
C) Interference
D) Mie Scattering
E) Refraction